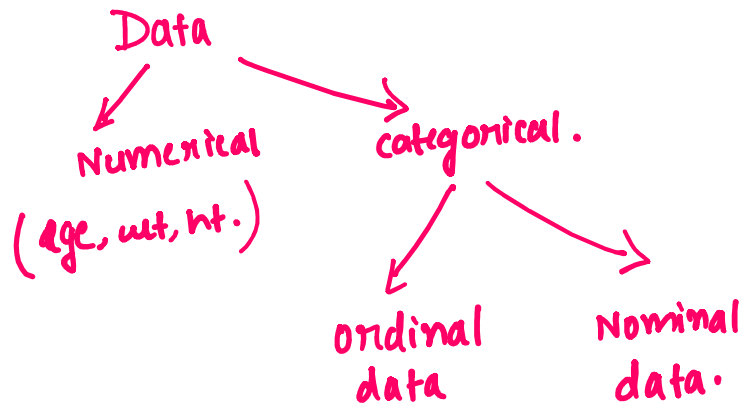


2 Handling categorical data



ordinal data :- categories which is having same mathematical / logical order.

Students

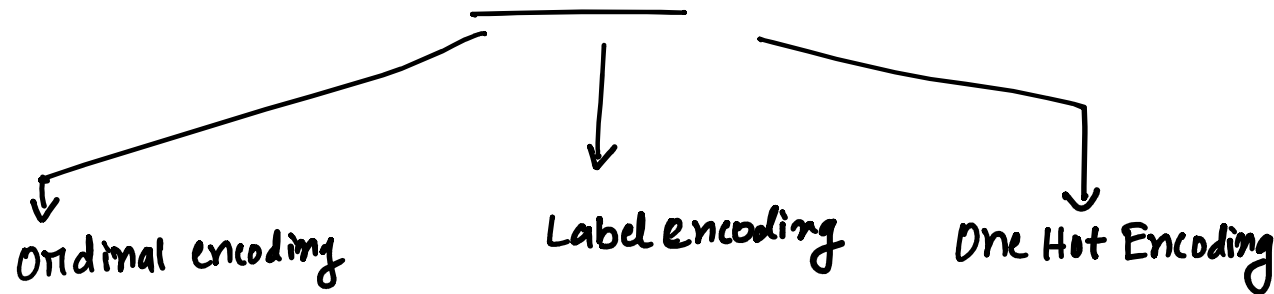
high school $\times$ UG < PG

Nominal data :- with no inherent orders.

states of India branches eng

Encoding :- Technique of transforming categorical data into numerical.

Three categories



1) Ordinal Encoding

Poor

(0)

Avg

(1)

Good

(2)

Python implementation

from sklearn.preprocessing import OrdinalEncoder

oe = OrdinalEncoder(categories=[['Poor', 'avg', 'good'],
['hs', 'un', 'pr']])

x_train_transformed = oe.fit_transform(x_train)

x_test_transformed = oe.fit_transform(x_test)

2

Label Encoding

email

cancer

email

Spam

Not Spam

cancer

Yes

NO

from sklearn.preprocessing import LabelEncoder.

le = LabelEncoder()

x_train_transformed = le.fit_transform(x_train)

x_test_transformed = le.fit_transform(x_test)

3 OneHot Encoding → Nominal data

colours

Red ✓
Yellow ✓
blue ✓

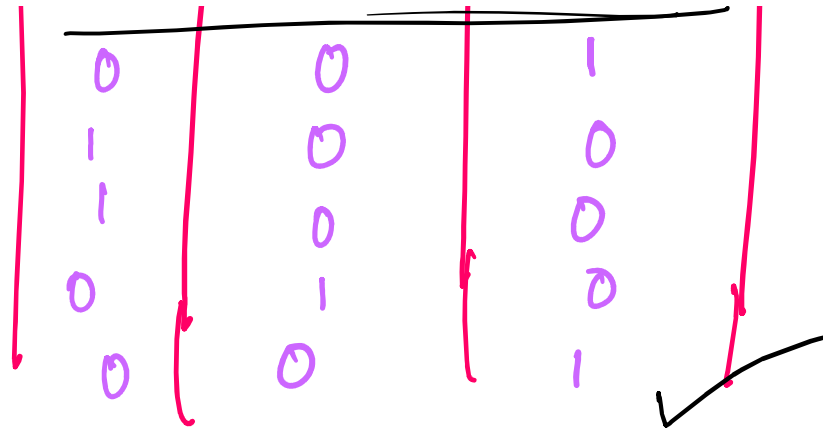
no orders.

Red	Yellow	Blue
1	0	0
0	1	0
0	0	1

Sparse matrix

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

blue ✓
 red
 red
 yellow
 blue.



Multi collinearity

$$\sum x_i = 1$$

$$[1 \ 0 \ 0] = 1$$

$$[0 \ 0 \ 1] = 1$$

color

Red

Yellow

Red

1

0

Yellow

0

1

Blue

0

0

red

1

0

red

1

0

Yellow.

0

1